

In [1]:

```
import sqlite3
import pandas as pd

conn = sqlite3.connect("ev_project.db")
cursor = conn.cursor()
```

In [2]:

```
# Load the Cleaned DataFrames
df_flat_cleaned = pd.read_pickle("df_flat_cleaned.pkl")
df_web_cleaned = pd.read_pickle("df_web_cleaned.pkl")
df_api_cleaned = pd.read_pickle("df_api_cleaned.pkl")

print(df_flat_cleaned.head())
print(df_web_cleaned.head())
print(df_api_cleaned.head())
```

	region	category	parameter	mode	powertrain	year	unit	value
2	Australia	Historical	EV sales	Cars	BEV	2011	Vehicles	49.0
3	Australia	Historical	EV stock	Cars	BEV	2011	Vehicles	49.0
4	Australia	Historical	EV stock	Cars	BEV	2012	Vehicles	220.0
5	Australia	Historical	EV sales	Cars	BEV	2012	Vehicles	170.0
8	Australia	Historical	EV stock	Cars	PHEV	2012	Vehicles	80.0

		Car Model	Base Price (USD)	\
0	2024	Acura Zdx A-Spec Rwd 20-In	64500.0	
1	2024	Acura Zdx A-Spec Awd 20-In	68500.0	
2	2024	Acura Zdx Type S Awd 22-In	73500.0	
3	2024	Acura Zdx Type S (Perf. Tires) Awd 22-In	74500.0	
4	2024	Audi E-Tron Gt 20-In	106500.0	

	Destination Charge (USD)	Tax Credit (USD)	Effective Price (USD)
0	1350.0	NaN	65850.0
1	1350.0	NaN	69850.0
2	1350.0	NaN	74850.0
3	1350.0	NaN	75850.0
4	1495.0	NaN	107995.0

	Station Name	Town	State	\
0	Ionna	San Francisco	CA	
1	Aloft Arlington	Arlington	TX	
2	Smart & Final	Montebello	CA	
3	Bishop Ranch 3	San Ramon	CA	
4	Pilot Travel Center #1174	Marshall, IL	Marshall	IL

	Number of Points	Usage	Type	Operator	Status	Latitude	Longitude
0	16	None	None	None	None	37.717485	-122.442329
1	1	None	None	None	None	32.761525	-97.075392
2	68	None	None	None	None	34.030711	-118.126120
3	20	None	None	None	None	37.764959	-121.958355
4	4	None	None	None	None	39.418436	-87.694053

Out[3]:

10

In [4]:

```
# ev_flat, ev_web, and ev_api are at different levels of granularity and
# don't share a real join key, so each is queried independently below.

query_usa_sales = """
SELECT region, year, parameter, value
FROM ev_flat
WHERE LOWER(region) = 'usa'
ORDER BY year
"""

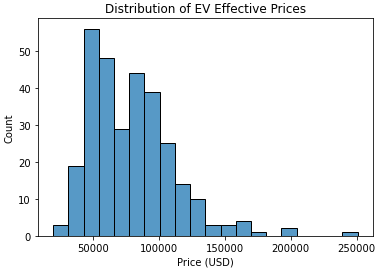
usa_sales = pd.read_sql_query(query_usa_sales, conn)
print(usa_sales.head())
```

	region	year	parameter	value
0	Usa	2010	EV sales	1200.0
1	Usa	2010	EV stock	3800.0
2	Usa	2011	EV stock	14000.0
3	Usa	2011	EV sales	9800.0
4	Usa	2011	EV stock	8000.0

In [5]:

```
import matplotlib.pyplot as plt
import seaborn as sns

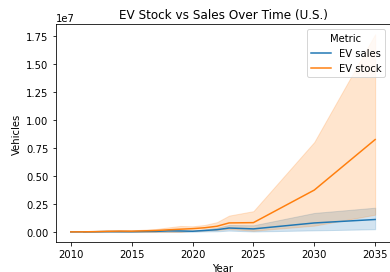
sns.histplot(data=df_web_cleaned, x="Effective Price (USD)", bins=20)
plt.title("Distribution of EV Effective Prices")
plt.xlabel("Price (USD)")
plt.ylabel("Count")
plt.show()
```



In [6]:

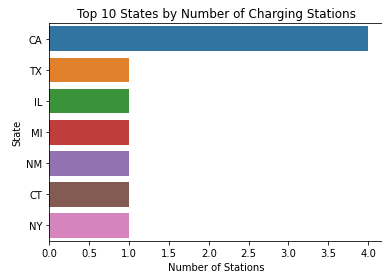
```
df_sales_stock = df_flat_cleaned[df_flat_cleaned['region'].str.lower() == 'usa']

sns.lineplot(data=df_sales_stock, x='year', y='value', hue='parameter')
plt.title("EV Stock vs Sales Over Time (U.S.)")
plt.xlabel("Year")
plt.ylabel("Vehicles")
plt.legend(title="Metric")
plt.show()
```



```
In [7]: station_counts = df_api_cleaned['State'].value_counts().head(10).reset_index()
station_counts.columns = ['State', 'Count']

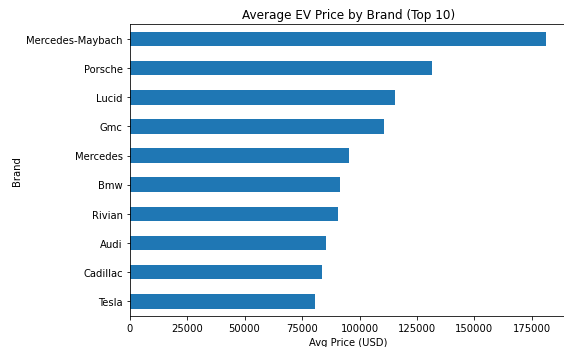
sns.barplot(data=station_counts, x='Count', y='State')
plt.title("Top 10 States by Number of Charging Stations")
plt.xlabel("Number of Stations")
plt.ylabel("State")
plt.show()
```



```
In [8]: # Extract the brand as the first word of the Car Model field
df_web_cleaned['Brand'] = df_web_cleaned['Car Model'].str.split().str[1]
# (First word is usually the model year, e.g. "2024" - second word is the brand)

avg_price_by_brand = (
    df_web_cleaned.groupby('Brand')['Effective Price (USD)']
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

plt.figure(figsize=(8, 5))
avg_price_by_brand.plot(kind='barh')
plt.title("Average EV Price by Brand (Top 10)")
plt.xlabel("Avg Price (USD)")
plt.ylabel("Brand")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



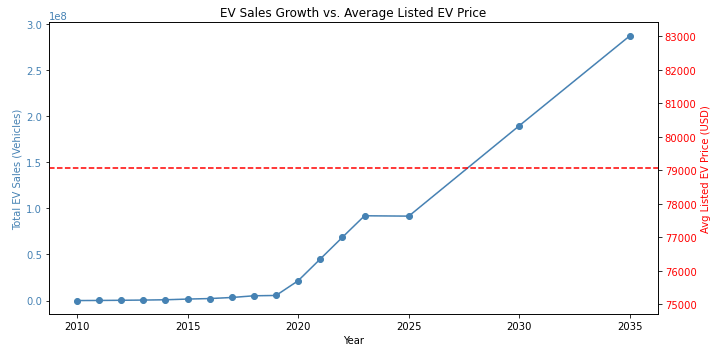
```
In [9]: avg_price = df_web_cleaned['Effective Price (USD)'].mean()
total_sales = (
    df_flat_cleaned[df_flat_cleaned['parameter'] == 'EV sales']
    .groupby('year')['value'].sum().reset_index()
)

fig, ax1 = plt.subplots(figsize=(10, 5))

ax1.plot(total_sales['year'], total_sales['value'], marker='o', color='steelblue', label='EV Sales')
ax1.set_xlabel("Year")
ax1.set_ylabel("Total EV Sales (Vehicles)", color='steelblue')
ax1.tick_params(axis='y', labelcolor='steelblue')

ax2 = ax1.twinx()
ax2.axhline(avg_price, color='red', linestyle='--', label='Avg Listed EV Price')
ax2.set_ylabel("Avg Listed EV Price (USD)", color='red')
ax2.tick_params(axis='y', labelcolor='red')

plt.title("EV Sales Growth vs. Average Listed EV Price")
fig.tight_layout()
plt.show()
```



```
In [10]: # Summary of Findings:

# 1. EV Sales & Stock (ev_flat): U.S. EV stock and sales trend upward, with stock growing faster than annual sales.

# 2. EV Listing Prices (ev_web): Prices are right-skewed, mostly clustering between $50k-$100k.

# 3. Charging Infrastructure (ev_api): Stations are concentrated in a few states, led by California.

# 4. Data Limitation: The three sources don't share a common key, so each was analyzed independently rather than joined.

In [ ]:
```